

Schematic configuration of the ABAC symporter transport mechanism,
where the order of binding is A, B, second A, and C
and the unbinding reactions occur in the reverse order of binding

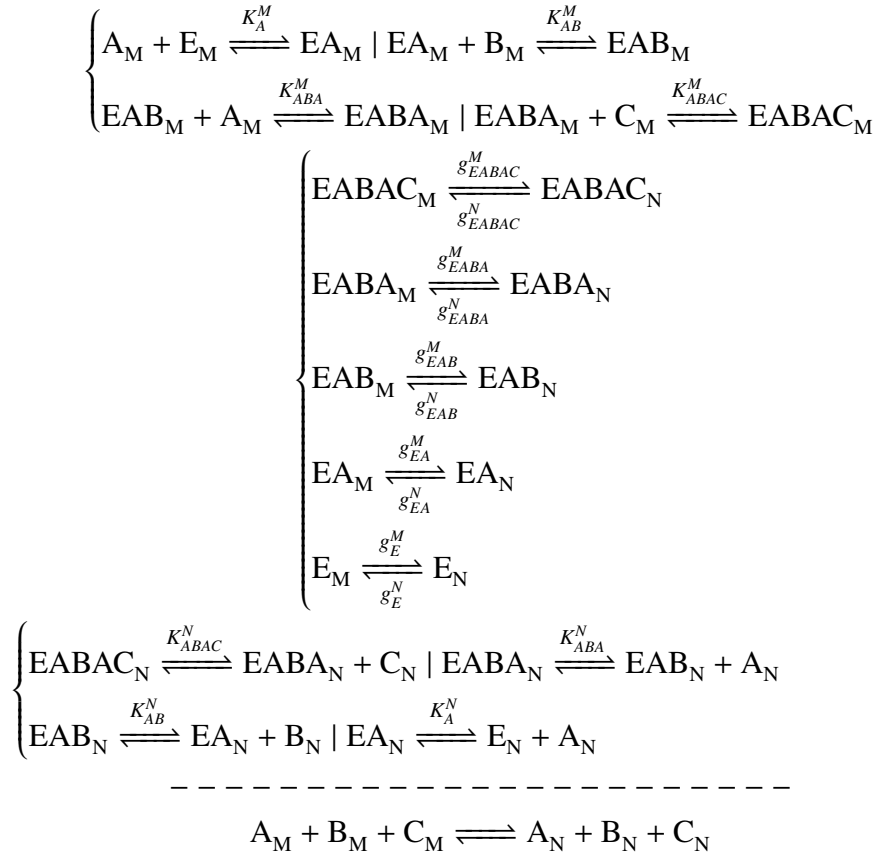
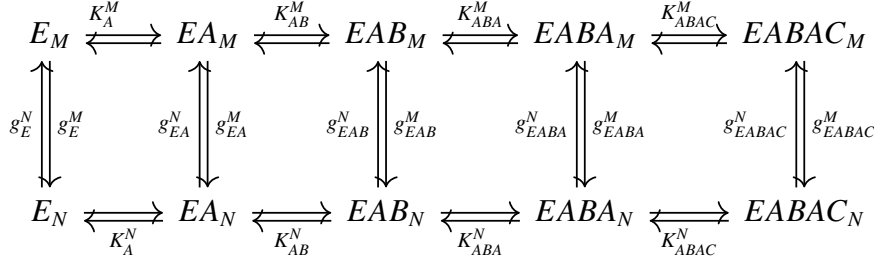


Figure 14: A schematic configuration of the ABAC symporter mechanism transporting of three solutes and the regarding net flux transport equation. In this configuration, the carrier exists in 5 different states on each side of the membrane. Dissociation constants (K_A , K_{AB} , K_{ABA} , and K_{ABAC} terms) and the translocation rate constants (g_E , g_{EA} , g_{EAB} , g_{EABA} , and g_{EABAC} terms) are depicted in the reaction cycle in the right panel. $[E]_t$ is the total number of transporter per unit area of the membrane. Transport equations are reparametrized and transformed into the general resistance form from Delpire model [134] (part 1/2 continued on next page).

Net Transport Flux from M to N

$$J_{A, \text{symporter}}^{M,N(\text{net})} = [E]_t \left(\frac{R_{NN} (g_{EA}^M \alpha^M + g_{EAB}^M \alpha^M \beta^M + g_{EABA}^M \alpha^M \beta^M \alpha''^M + g_{EABAC}^M \alpha^M \beta^M \alpha''^M \gamma^M)}{R_M R_{NN} + R_N R_{MM}} - \frac{R_{MM} (g_{EA}^N \alpha^N + g_{EAB}^N \alpha^N \beta^N + g_{EABA}^N \alpha^N \beta^N \alpha''^N + g_{EABAC}^N \alpha^N \beta^N \alpha''^N \gamma^N)}{R_M R_{NN} + R_N R_{MM}} \right) \quad (58a)$$

$$J_{B, \text{symporter}}^{M,N(\text{net})} = [E]_t \left(\frac{R_{NN} (g_{EAB}^M \alpha^M \beta^M + g_{EABA}^M \alpha^M \beta^M \alpha''^M + g_{EABAC}^M \alpha^M \beta^M \alpha''^M \gamma^M)}{R_M R_{NN} + R_N R_{MM}} - \frac{R_{MM} (g_{EAB}^N \alpha^N \beta^N + g_{EABA}^N \alpha^N \beta^N \alpha''^N + g_{EABAC}^N \alpha^N \beta^N \alpha''^N \gamma^N)}{R_M R_{NN} + R_N R_{MM}} \right) \quad (58b)$$

$$J_{C, \text{symporter}}^{M,N(\text{net})} = [E]_t \left(\frac{R_{NN} (g_{EABAC}^M \alpha^M \beta^M \alpha''^M \gamma^M) - R_{MM} (g_{EABAC}^N \alpha^N \beta^N \alpha''^N \gamma^N)}{R_M R_{NN} + R_N R_{MM}} \right) \quad (58c)$$

where

$$[E]_t = [E]_M + [EA]_M + [EAB]_M + [EABA]_M + [EABAC]_M + [EABAC]_N + [EABA]_N + [EAB]_N + [EA]_N + [E]_N$$

$$\begin{aligned} \alpha^M &= \frac{[A]_M}{K_A^M}, \beta^M = \frac{[B]_M}{K_{AB}^M}, \alpha''^M = \frac{[A]_M}{K_{ABA}^M}, \gamma^M = \frac{[C]_M}{K_{ABAC}^M} \\ \alpha^N &= \frac{[A]_N}{K_A^N}, \beta^N = \frac{[B]_N}{K_{AB}^N}, \alpha''^N = \frac{[A]_N}{K_{ABA}^N}, \gamma^N = \frac{[C]_N}{K_{ABAC}^N} \\ R_M &= 1 + \alpha^M + \alpha^M \beta^M + \alpha^M \beta^M \alpha''^M + \alpha^M \beta^M \alpha''^M \gamma^M \\ R_N &= 1 + \alpha^N + \alpha^N \beta^N + \alpha^N \beta^N \alpha''^N + \alpha^N \beta^N \alpha''^N \gamma^N \\ R_{MM} &= g_E^M + g_{EA}^M \alpha^M + g_{EAB}^M \alpha^M \beta^M + g_{EABA}^M \alpha^M \beta^M \alpha''^M + g_{EABAC}^M \alpha^M \beta^M \alpha''^M \gamma^M \\ R_{NN} &= g_E^N + g_{EA}^N \alpha^N + g_{EAB}^N \alpha^N \beta^N + g_{EABA}^N \alpha^N \beta^N \alpha''^N + g_{EABAC}^N \alpha^N \beta^N \alpha''^N \gamma^N \end{aligned}$$

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Figure 15 depicts a different binding/unbinding order for multiple solutes on a carrier. In the scheme, the stoichiometric ratio of the transport reaction is $A : 2B : C$ and the order of binding is as follows: first, solute A binds to the carrier on surface M of the membrane to form the complex EA_M